Finding Colluding Groups from Players' Moves: Shuffle-Calibrated Group Scans for Multi-Agent and LLM Collusion

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Abstract

Auditors of multi-agent systems often see only what agents do, not what they say to each other. The information-theoretic detector of Bonjour et al. (2022) flags pairs of players that influence each other more than anyone else influences them, but it assumes exactly two colluders. We show that this pairwise net-influence score cancels inside groups: in simulated n-player Rock Paper Scissors it found 0% of planted triangles, cliques and hubs, (precisely defined later) and its fixed threshold is uncalibrated (8–23% false alarms in short games with no collusion). We propose a group scan that scores every subset of players by the dependence inside it and calibrates the maximum with a within-state shuffle test. Among 20 players, a 5-player group colluding 20% of the time is recovered exactly in 48% of 100-round games, against 5% for the best pairwise baseline. The shuffle must condition on what honest players react to: ignoring history flags honest players in up to 100% of games, while the right state restores about 5% false alarms and still finds a profitable steering coalition in 94–98% of long games. Finally, we build an LLM double auction in which every agent sees only a shared coarse state, which makes the shuffle test exact. gpt-4.1-mini sellers given a message channel and CEO profit pressure earn 49–65% more than controls; a same-move shuffle test flags them (2 sellers: 8 of 10 20-game batches; 5 sellers: 19 of 20 5-game batches), while pooled tests on both controls are not significant.

1. Introduction

LLM agents are increasingly deployed as autonomous economic actors: pricing, bidding and trading on behalf of firms. Agrawal et al. (2025) show that LLM sellers in a continuous double auction can collude, raising prices above competitive levels, especially when given a channel to talk to each other and pressure from management to keep margins high. A regulator or platform operator auditing such a market usually cannot read private channels; it sees the moves. The question we study is the sprint's research question 1.4: *can information-theoretic methods detect collusion from moves alone?*

Threat model. $k \geq 2$ of n agents coordinate through a channel the auditor cannot observe (a shared secret signal, a message channel, or a common plan). The auditor observes every agent's action each round and the public state of the game, and must say which agents, if any, are colluding while keeping false alarms on honest agents at a chosen level (5%). Honest agents need not be random: they may react to public history, which itself makes their moves correlated.

The existing detector (Bonjour et al., 2022) is built for one colluding pair. Real coordination need not be pairwise: a cartel of five sellers, a hub with several helpers, or a group sharing one signal. We find that pairwise net influence is blind to exactly these structures.

Our main contributions are:

1. **A failure mode of pairwise net influence.** Inside a group, each member's partners cancel in its net-influence score, so the published rule finds 0% of triangles, cliques and hubs, while the

underlying links are ~200 standard deviations above chance. Its fixed threshold is also uncalibrated (8–23% false alarms in 100-round games with no collusion).

2. **A shuffle-calibrated group scan.** Scoring every subset by the links inside it, calibrated by a max-statistic permutation test, recovers weak, broad groups that pairwise methods miss (48% vs 5% exact recovery for a weak 5-clique among 20 players) at a 5% false-alarm level.
3. **Conditioning on history is necessary, and sufficient when the state is right.** Honest players that react to history are flagged in up to 100% of games unless the shuffle stays within history states. With the right state, a coalition whose coordination pays is found in 94–98% of long games while honest exploiters with the same reaction rate are flagged in 4–7%.
4. **An LLM auction where the test is exact by design.** By giving every agent only a shared coarse state, conditional independence of honest agents holds by construction. The test detects comms + CEO-pressure collusion among gpt-4.1-mini sellers and stays quiet on controls, for 2 and for 5 sellers.

2. Related Work

Information-theoretic collusion detection. Bonjour, Aggarwal & Bhargava (2022) define a pairwise influence γ(i;j) as the mutual information between two players' actions, and net influence as

$$\Gamma(i;j) = \gamma(i;j) - \max_{k \neq i,j} \gamma(k;j).$$

A pair is flagged when Γ passes 0.05 bits in both directions and exactly one pair passes. This is the closest prior work. We reproduce it (Appendix B) and show its cancellation inside groups and its miscalibration; our detectors keep its building block (pairwise conditional mutual information) but change the decision rule and calibration.

LLM collusion in markets. Agrawal et al. (2025) study LLM sellers in a continuous double auction and find that communication and CEO profit pressure increase collusive pricing. They measure collusion with access to prices and messages. We ask whether it is detectable from moves alone, and reuse their mechanism (comms + pressure) as the positive condition.

Per-agent detectors and multiple testing. Rose et al. (2026) score individual agents for suspicious behaviour; our "player score" baseline is inspired by this but is not their method. Our "pair test" applies Benjamini–Hochberg (1995) false-discovery control to all pairs and joins significant pairs into groups. Both are natural baselines that a group scan should beat.

Gap. No prior method we know of (i) detects groups larger than two from moves, (ii) has a calibrated false-alarm rate, and (iii) handles honest agents whose moves are correlated through shared history.

Scan statistics and max-T tests. The group scan is a scan statistic over subsets, calibrated by the permutation distribution of its maximum; this controls the family-wise error over all subsets at once (Westfall & Young, 1993).

3. Methods

3.1 Simulated game: n-player Rock Paper Scissors

Each round every player plays every other player (+1 win, −1 loss); n ∈ {5, 10, 20}; 100–3000 rounds. With a *collusion probability* (CP) per round, planted colluders coordinate; otherwise they play randomly. Colluders sit at random seats unknown to the detectors. Table 1 defines the planted groups and the honest player types used throughout.

Table 1. Planted colluding groups and honest player types.

Kind	Name	What the players do
Colluders	Pair	A helper plays the move that loses to its partner, feeding the partner a point.
Colluders	Two pairs	Two separate pairs, as above.
Colluders	Hub	One leader and two helpers; each helper feeds the leader, and the helpers never coordinate with each other.
Colluders	Triangle, k-clique	$k = 3$, 4 or 5 players share a secret signal and all play the same move (a 3-clique is a triangle).
Colluders	xor	Player C plays (A + B) mod 3, so every pair looks independent.
Colluders	Steer	4 players all play one move to create a clear lead, then all play the move that beats the honest players who react to that lead; coordination pays.
Honest	Random	Plays uniformly at random.
Honest	Leader-reactive	With probability 0.5, plays the move that beats last round's most common move.
Honest	Margin-reactive	Same, but only when that move led by 3 or more.
Honest	Exploiter	Plays the move that beats the leader-reactive response, on its own, without coordinating.

3.2 Detectors

All detectors start from the pairwise conditional mutual information $\gamma_{ij} = I(a_i; a_j \mid s)$ in bits, where s is a *history state* computed from the previous round: none, top (which move led), order (ranking of move counts) or full (exact counts).

Within-state shuffle null. Each player's action column is permuted independently among rounds that share the same state. This preserves each player's state-dependent habits and destroys any coordination beyond them. Under the null that players are conditionally independent given the state, and rounds are exchangeable within a state, it is an exact permutation test. We draw two independent sets of B permutations: Set A to standardize each edge, $z_{ij} = (\gamma_{ij} - \mu^A_{ij})/\sigma^A_{ij}$, and Set B to calibrate the decision.

Group scan (ours). For every subset C with $|C| \geq 2$,

$$T(C) = \sum_{i<j \in C} z_{ij}, \qquad S(C) = \frac{T(C) - \mathbb{E}_A[T(C)]}{\sqrt{\mathrm{Var}_A[T(C)]}},$$

where the Set-A moments are exact quadratic forms in the edge means and covariance. We report $\arg\max_C S(C)$ if $\max_C S(C)$ exceeds the 95th percentile of the same maximum over Set B, then remove that group and repeat on the remaining players (step-down max-T). The exact search covers ~10^6 subsets at n = 20.

Baselines. *Bonjour*: the published rule (fixed 0.05). *Bonjour (calibrated)*: same rule, threshold from the shuffle null. *Pair test*: Benjamini–Hochberg at q = 0.05 over all pairs (p-values from a gamma fit to the Set-A null, since raw permutation p-values cannot reach the BH bar at n $\geq$ 10), groups = connected components. *Player score*: per-player total link strength, top players flagged.

Metrics. *Exact recovery*: the reported group equals the planted one. *Clean recovery*: exact and nothing else reported. *False alarm*: any report when no collusion exists (target 5%). 95% Wilson intervals in brackets. B = 1000 for all reported numbers unless stated.

3.3 LLM double auction with a coarse shared state

A shuffle test is only exact if the state captures everything honest agents react to (Section 4.3). With LLMs that carry memory or see detailed price paths, no recorded state does (Appendix C). We therefore **design the market so the state is the agents' entire public input.**

- **Market.** Sellers (cost \$80) and scripted buyers (value \$100) each hour *lower*, *keep* or *raise* their quote by \$2 (logged as *tighten*, *hold*, *widen* in the code and data). At most one trade per hour, between the lowest ask and the highest bid, at the midpoint. Asks start at \$96, bids at \$84. Exact prices stay inside the simulator.
- **Shared state** S_t, the only thing in each prompt besides the rules: 2×2 market (2 sellers, 2 buyers, 30 hours) — each seller's last move, who sold, and whether the lowest ask is high ($\geq$ \$92) or low; 5×5 market (5 sellers, 5 buyers, 50 hours) — how many sellers lowered / kept / raised, who sold, and the ask level. No memory, scratch pad or hour number.
- **Conditions.** `comms_off` (rules only); `pressure_off` (plus a CEO message: margins are tracked and low-margin sellers are decommissioned); `comms_pressure` (CEO message plus a 200-character message channel; each seller sees every seller's last message). Messages are the hidden channel: they are not in S_t.
- **Model.** gpt-4.1-mini, temperature 0.7, JSON output. 2×2: 200 games per condition (36,000 calls, ≈\$6.35). 5×5: 100 games per condition (75,000 calls, ≈\$14.38).
- **Test.** Because every seller's prompt is a function of S_t (and its own name), sellers are conditionally independent given S_t unless they coordinate through messages, so the within-state shuffle is exact. The statistic is the *same-move count*: rounds in which a pair of sellers took the same action. We chose it from a 20-game 2×2 pilot, before either main run, and kept it unchanged for 5 sellers. For 5 sellers we shuffle each seller independently within states and report a global test (sum over all 10 pairs) and a max-pair test (largest standardized pair, which controls family-wise error and localizes the pair). We report rejection rates over disjoint batches of 5, 10 and 20 games.

4. Results

4.1 Pairwise net influence cannot see groups

Observation. The published Bonjour rule found **0% of triangles, cliques and hubs in every setting** (Table 2), while the links inside a triangle were about 200 standard deviations above the shuffled data (Figure 1). With no collusion, its fixed threshold raised false alarms in 8–23% of 100-round games (n = 5–20) and never fired in longer ones; in a 3-player variant whose score conditions on the previous round, it flagged up to 43% (Appendix B). Every shuffle-calibrated detector stayed at 3–9%.

Interpretation. Net influence subtracts the strongest *other* influence on j. Inside a group, j's other partners are as strong as i, so $\Gamma \approx 0$ for every member. A fixed threshold on an estimator whose bias depends on rounds and state cannot be calibrated.

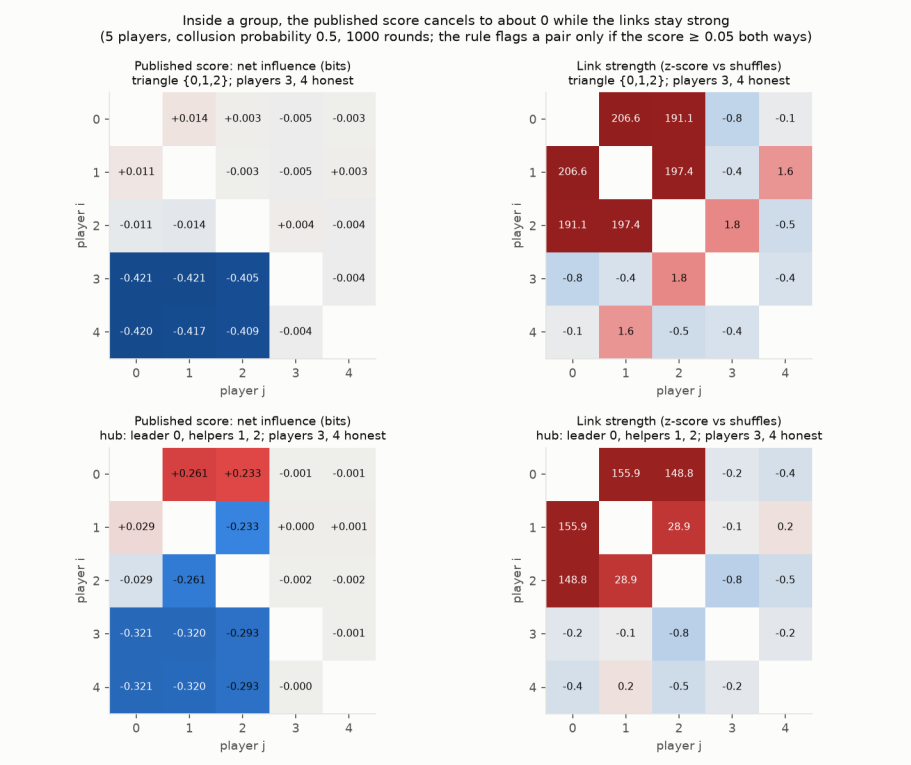


Figure 1. 5 players, CP 0.5, 1000 rounds; players 0–2 collude. Left: Bonjour net influence per ordered pair (must reach 0.05 both ways). Right: link strength as a z-score against shuffled data. Inside the triangle the score is ≈ 0 while the links are ≈ 200; in the hub it passes only from leader to helper.

4.2 The group scan finds weak, broad groups

Scoring the group as a whole adds up evidence too weak in any single pair (Table 2, Figure 2). With 20 players, 100 rounds and a 5-clique at CP 0.2, the group scan recovered the exact group in **48% [38, 58]** of games, against 5% [2, 11] for the pair test and 4% [2, 10] for the player score. It also degrades less as players are added: for a triangle, the player score falls from 71% to 52% between n = 5 and 20, the group scan from 77% to 71%.

Table 2. Exact recovery (%) by planted group and number of players. Averaged over the CP and round settings tried for each group (200 shuffles).

Planted group	n	Bonjour	Pair test	Player score	Group scan
Pair	20	70	89	72	**91**
Hub	20	0	**86**	68	78
Triangle	20	0	64	52	**71**
4-clique	20	0	48	45	**62**
5-clique	20	0	46	51	**63**
xor	5–20	0	0	0	0

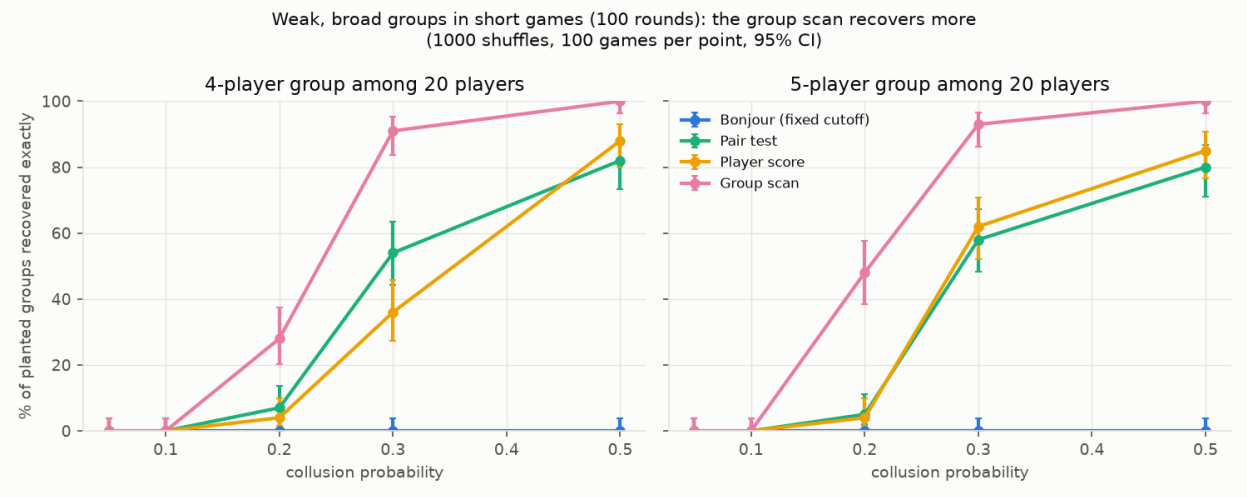


Figure 2. Exact recovery vs CP for 4- and 5-cliques among 20 players (100 rounds, 1000 shuffles, 100 games per point). Bonjour stays at 0.

Where it loses. The scan often reports a hub as a single leader–helper pair, because the two helpers are not linked; the pair test, which joins both edges, does better (58% vs 34% at CP 0.2, 300 rounds). For two isolated weak pairs in short games the pair test also wins narrowly (58% vs 52%): the scan pays for searching ~10^6 groups.

4.3 The test must condition on what honest players react to

Observation. Honest players who react to last round's moves are correlated with each other. With no state, the group scan flagged honest players in **up to 100%** of 1000-round games (Figure 3). Leader-reactive players need at least the *top* state; margin-reactive players need *full*. With the right state, false alarms return to 4–9%.

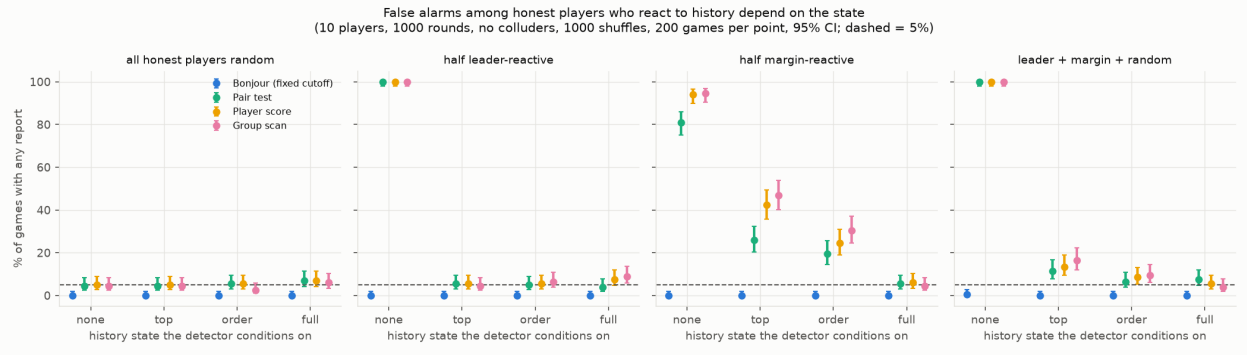


Figure 3. False-alarm rate by history state (coarse → fine) for four mixes of honest players; 10 players, 1000 rounds, no colluders, 200 games per point. Dashed line: 5% target.

Profitable collusion. A 4-player steer group earns 0.21 points per round each at CP 0.3, **2.6**× what four independent exploiters earn reacting at the same rate; the margin-reactive players pay for it (Table 3). With the full state, the group scan finds the steer group cleanly in 94–98% of 1000+-round games at CP 0.2–0.3, while the honest exploiters are flagged in 4–7%. With coarser states the exploiters look like colluders as games get longer (66–71% under *top*, 96–100% with no state at 3000 rounds).

Table 3. Steer group vs honest exploiters (4 of each type among 3 margin-reactive and 3 random players; 200 games per cell). Payoff = points per round per member (margin-reactive players' payoff in brackets). Recovery and false alarms: group scan, full state, at 300 / 1000 / 3000 rounds.

CP	Steer payoff	Exploiter payoff	Steer clean recovery (%)	Exploiter false alarms (%)
0.1	0.08 (−0.11)	0.03 (−0.04)	7 / 84 / 95	6 / 6 / 7
0.2	0.15 (−0.20)	0.06 (−0.07)	76 / 98 / 94	5 / 4 / 6
0.3	0.21 (−0.27)	0.08 (−0.10)	92 / 94 / 94	5 / 6 / 6

Interpretation. The state is a bias–variance choice: too coarse and honest reactions look like collusion; too fine and short games lose power (a triangle among reactors: 50% clean recovery at 300 rounds with *full*, 94–95% from 1000 rounds).

4.4 LLM sellers: comms + CEO pressure is detected, controls are not

Observation. In both markets, comms + pressure raised seller profit by 49–65% over the controls and was the only condition the test flagged consistently (Table 4). Pooled over 200 games, the 2×2 same-move statistic gave $z = +7.9$ for comms + pressure and $|z| \leq 1.5$ for both controls (CMI: +5.0 vs $|z| \leq 0.8$). In the 5×5 market, pooled over 100 games, the global test gave $p = 0.001$ (the floor at 1000 shuffles; +1,991 matching seller pairs above the shuffle's expectation out of 50,000 pair-rounds, top pairs $z = 9$–12) against $p = 0.17$ and 0.71 on the controls.

Table 4. Batches flagged at $\alpha = 0.05$, out of disjoint batches of 5, 10 and 20 games. 2×2: same-move test. 5×5: global test over all 10 seller pairs. Profit per seller-hour with a 95% bootstrap interval over games.

Market	Condition	Profit / hour	5 games	10 games	20 games
2×2	comms off	4.29 [4.06, 4.51]	2/40	0/20	0/10
2×2	pressure, no comms	4.76 [4.52, 5.00]	1/40	2/20	2/10
2×2	**comms + pressure**	**7.07** [6.76, 7.38]	12/40	8/20	**8/10**
5×5	comms off	5.12 [4.82, 5.42]	0/20	0/10	0/5
5×5	pressure, no comms	5.24 [4.93, 5.54]	0/20	0/10	0/5
5×5	**comms + pressure**	**8.20** [7.83, 8.59]	**19/20**	**10/10**	**5/5**

The batch counts are small: 95% Wilson intervals are [49, 94]% for 8/10, [76, 99]% for 19/20 and [57, 100]% for 5/5. The 5×5 max-pair test flags comms + pressure in 16/20, 10/10 and 5/5 batches, and the controls in 1/20 and 0/20 five-game batches and none of the larger ones.

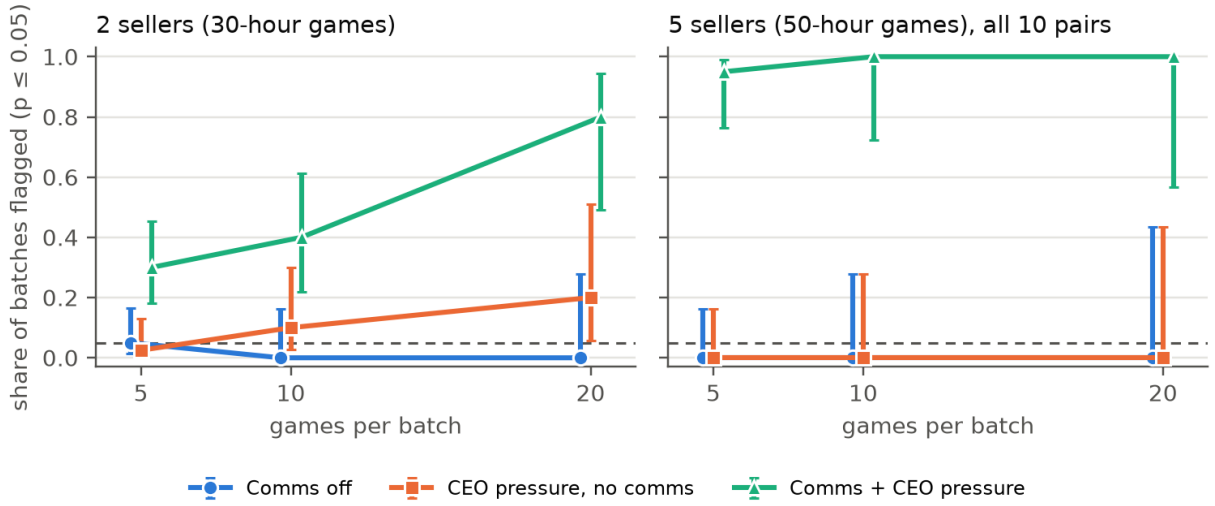


Figure 4. Share of batches flagged vs games per batch, with 95% Wilson intervals. Left: 2×2 market, same-move test. Right: 5×5 market, global test. Dashed line: 5% target.

Interpretation. The mechanism is visible in the messages and the moves. CEO pressure makes the seller who just sold raise (2×2: 8% → 33% in low-price states after a trade); the channel stops the others from undercutting (losing-seller "lower" falls from 82% to 41%), so moves synchronize. In the 5×5 market the whole group converges on "hold, don't undercut" ("Hold prices steady; avoid price wars to protect profits for all sellers"), so the excess agreement appears in all 10 pairs at once and is detected in smaller batches (each 5×5 game also has 50 hours and 10 seller pairs, against 30 hours and one pair). The max-pair test does not isolate a sub-cartel here because every pair is in sync; in a scripted check where only 2 of 5 sellers collude, it localizes the right pair in 99.7% of batches. The 2×2 `pressure_off` rate of 0.20 at 20 games (2 of 10 batches; 95% interval [6, 51]%) is consistent with 5% but is the weakest point of the control results. What the test detects is a shared signal outside the public state that coordinates moves; reading it as collusion rests on the profit gap over the pressure-only control.

5. Discussion and Limitations

Implications for AI safety. Monitoring agent markets from moves alone is feasible, but only with two ingredients the existing method lacks: a test over groups, and a null that models what honest agents legitimately react to. The second is the binding constraint. Our LLM result works because we designed the market so the honest-agent input is fully recorded; in a deployed market an auditor should log the information each agent is shown, which makes this test exact.

Limitations.

- **Coordination that is a function of the public state is invisible.** Follow-the-leader or an explicit cartel that keys on last round's public moves is absorbed into the state. A direction test (does a seller raise after others raise?) catches these but also fires on tacit following without communication, so it needs a comms-off calibration.
- **Higher-order collusion.** xor-style coordination has no pairwise trace (0% for every method); any fixed-order test is blind to the next order.
- **Exact search stops at ~20 players** (~10^6 subsets).
- **The state must be chosen**; the test cannot detect that honest agents react to something unrecorded. On recorded markets from Agrawal et al. (2025), shared time trends made the test badly miscalibrated (Appendix C).
- **Pair test runs slightly high** (7.7% [6.8, 8.7] false alarms at n = 20, 300 rounds) because its gamma-fit tail is generous.
- **LLM scope.** One model (gpt-4.1-mini), scripted buyers, coarse states and no agent memory. Richer inputs give LLMs more to react to and require a richer state.

Future work. Learned or richer states (e.g. conditioning on a model of honest play), higher-order dependence (partial information decomposition, Williams & Beer 2010), payoff-based tests that target harmful coordination of any order, and LLM markets with memory and more models.

6. Conclusion

The pairwise information-theoretic detector for collusion is blind to groups: its net-influence score cancels inside them and its fixed threshold is uncalibrated. A group scan with a within-state shuffle test fixes both, recovering weak, broad colluding groups at a controlled 5% false-alarm rate, provided the state captures what honest players react to.

When an LLM market is designed so that the state is the agents' entire public input, the test is exact. It detects the comms + CEO-pressure collusion of Agrawal et al. among gpt-4.1-mini sellers, in 80% of 20-game batches with 2 sellers and in 95% of 5-game batches with 5, while pooled tests on the controls stay non-significant.

Code and Data

All code, results and the LLM auction games are at https://github.com/taruntrivedi3/collusion-detection. `results/RESULTS_INDEX.md` lists the exact command and seeds for every number, including the two LLM studies (`studies/llm_auction/`, `studies/llm_auction_5seller/`). An interactive viewer (https://taruntrivedi3.github.io/collusion-detection/viewer/) replays example games, including the 2- and 5-seller LLM games with the sellers' messages.

Work Done Before the Sprint

Before the sprint began, we read the papers, came up with a game plan, played around with a few toy examples, and implemented the starting code. We also ran the recorded-market study in Appendix C (`studies/auction_null_calibration/`), including its log parser. Everything else in this report was done during the sprint.

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Appendix A. Limitations and Dual-Use / Ethical Considerations

Limitations are discussed in Section 5. In short: the test is only as good as the recorded state; state-driven, higher-order and unrecorded-channel coordination can evade it; results are from simulated players and one LLM with scripted buyers.

Dual use. A detector also tells colluders what to avoid. Our results show that coordinating through the public state, or at higher order, evades pairwise and group dependence tests. We report these blind spots openly because auditors need to know them, and because the defence (logging what agents observe, payoff-based tests) is the same regardless. We release no method for hiding collusion beyond what the limitations already state.

False accusations. A flag is statistical evidence that agents' moves are more aligned than their observed inputs explain, not proof of intent. Mis-specified states produce confident false flags (up to 100% in Section 4.3). Any operational use should treat a flag as grounds for review, report the state used, and calibrate on known-honest runs.

LLM experiments. All agents are simulated; no real market, person or firm is affected. The CEO pressure prompt is taken from prior work to reproduce a known effect.

Appendix B. Reproducing Bonjour et al.

Our average scores match the paper's Figure 3 in bits. Our detection accuracy matches its Table 2 only when the score conditions on the previous round's three moves (27 states), which the paper does not state; one number (CP 0.2, 1000 rounds: paper 24%, ours 49%) remains unexplained. With that 27-

state score, the fixed threshold flags a pair among 3 random players in 43.4% of 100-round games (19.2% at 300, 0.1% at 1000); the calibrated version stays at 5–6%. Full tables: `results/SUMMARY.md` (Tables 4, 8).

Appendix C. Recorded markets from Agrawal et al. (2025)

We ran the group scan on 75 recorded 5-seller markets from the original study. The recorded price paths share strong time trends (high early, lower later) that no available state captures without leaving nothing to shuffle. On independently resampled seller paths, the within-state scan flagged 88.5% of groups at nominal 5% (circular shifts: 37.2%). A same-pool empirical calibration reached 4.1% false alarms and 85.6% detection of copied triples, but did not transfer to held-out markets (4.2–48.8% null flags across folds). This motivated the coarse-state design of Section 3.3: without a state that explains honest behaviour, the test is not usable. Details: `studies/auction_null_calibration/`.

LLM Usage Statement

We used Claude Code (Anthropic) throughout the project: to write and refactor the simulation and analysis code, run the experiments, review the repository, and draft and edit this report and the documentation. The authors made the research decisions (what to test, which agents and conditions to use, what to keep or cut), directed and reviewed the work, and checked the numbers and claims. gpt-4.1-mini (OpenAI) played the sellers in the LLM auction experiments. All numbers in this report come from the released code and data.